CSE4063 Fundamentals of Data Mining Project#2

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Problem Definition

Problem

The objective of this project is to analyze Olympic athlete data over 120 years to uncover patterns, trends, and insights related to performance, medals, and country participation. The problem involves applying models for clustering athletes, predicting medal counts, and identifying key features influencing performance.

Why It Matters

This analysis helps understand trends in Olympic history, including performance improvements, country contributions, and athlete characteristics, with applications in sports analytics.

Dataset

Information About The Dataset

Dataset Name: 120 Years of Olympic History - Athletes and Results
Source: Kaggle (120-years-of-olympic-history-athletes-and-results)
Description: This dataset contains detailed information about Olympic athletes, events, and medals won across multiple Olympic Games from 1896 to the present.

Dataset Details

Number of Instances: Includes 271116 records.
Number of Columns: Includes 15 columns such as Year, City, Athlete, Country, Sport, Event, Medal, etc.

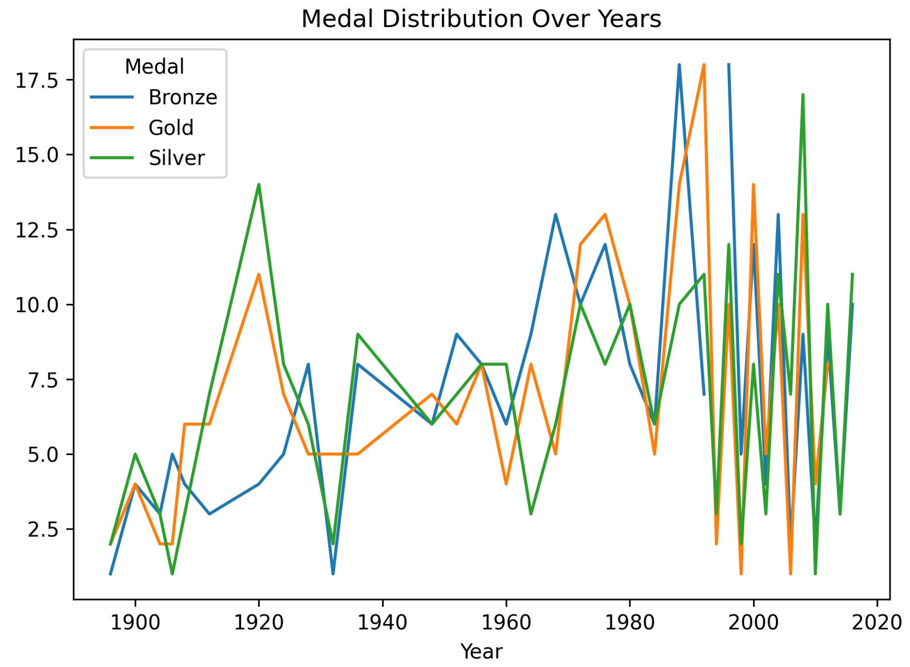
Dataset

Columns

ID - Unique number for each athlete;
Name - Athlete's name;
Sex - M or F;
Age - Integer;
Height - In centimeters;
Weight - In kilograms;
Team - Team name;
NOC - National Olympic Committee 3-letter code;
Games - Year and season;
Year - Integer;
Season - Summer or Winter;
City - Host city;
Sport - Sport;
Event - Event;
Medal - Gold, Silver, Bronze, or NA.

Dataset

Dataset Analysis



Data Preprocessing

Key Steps

- **Handling Missing Data:** To address data completeness, we implemented strategic imputation methods. For numerical features, we utilized median values to maintain central tendency without being affected by outliers. For categorical variables, we employed mode imputation to preserve the most frequent category distributions.
- **Categorical Variable Encoding:** We use OneHotEncoder to convert categorical variables into numerical format, creating binary columns for each category. This makes the data suitable for machine learning models.

Data Preprocessing

Key Steps

- **Numeric Feature Scaling:** StandardScaler is applied to normalize numeric features, ensuring all values are on the same scale. This prevents features with larger values from dominating the model training process.
- **Data Integration:** The processed numeric and categorical features are combined using hstack, creating a unified dataset that preserves both the transformed features and original identifying columns.

Implementation

Steps

- **Environment & Tools** — IDE/Environment and libraries used
- **Algorithms Implemented** — Details of the implemented algorithms
- **Implementation Details** — Details of the implementation

Implementation

Environment & Tools

- **IDE/Environment:** Code developed in Python using VSCode.
- **Libraries Used:**
    - **Data handling:** pandas, numpy.
    - **Preprocessing:** scikit-learn for OneHotEncoder and StandardScaler.
    - **Model Training:** scikit-learn for models including clustering (KMeans, DBSCAN), evaluation metrics (Silhouette Score, Calinski Harabasz Score, Davis Bouldin Score). Mlxtend for implementation of pattern mining algorithms (Apriori, FP-Growth)
    - **Visualization:** matplotlib, seaborn for visualizations.

Implementation

Models Implemented

- **Frequent Pattern Mining Models**
  - Apriori
  - FP-Growth
  - ECLAT
- **Clustering Analysis Methods**
  - K-Means
  - AGNES
  - DBSCAN

Implementation

Implementation Details

- **Preprocessing Pipeline:**
    - Handles missing values, scaling, and encoding features.
- **Model Creation:**
    - Defined in models.py using a dictionary structure for flexible model creation.

Implementation

Implementation Details

- **Training and Evaluation:**
  - Models are executed in main.py.
  - Logger class in logger.py saves runtime logs into text files for easy review.
  - Pattern mining and clustering evaluations are performed separately in evaluation.py.
  - The visualizations are created in visualization.py.

Model Evaluation & Performance Results

Here are the key findings from the model evaluation results:

1. Clustering Performance:

- K-means achieved moderate cluster separation with a Silhouette Score of 0.2703
- AGNES showed poorer separation with a lower Silhouette Score of 0.2343
- DBSCAN performed best with a Silhouette Score of 0.5462, suggesting better-defined clusters

2. Pattern Mining Results:

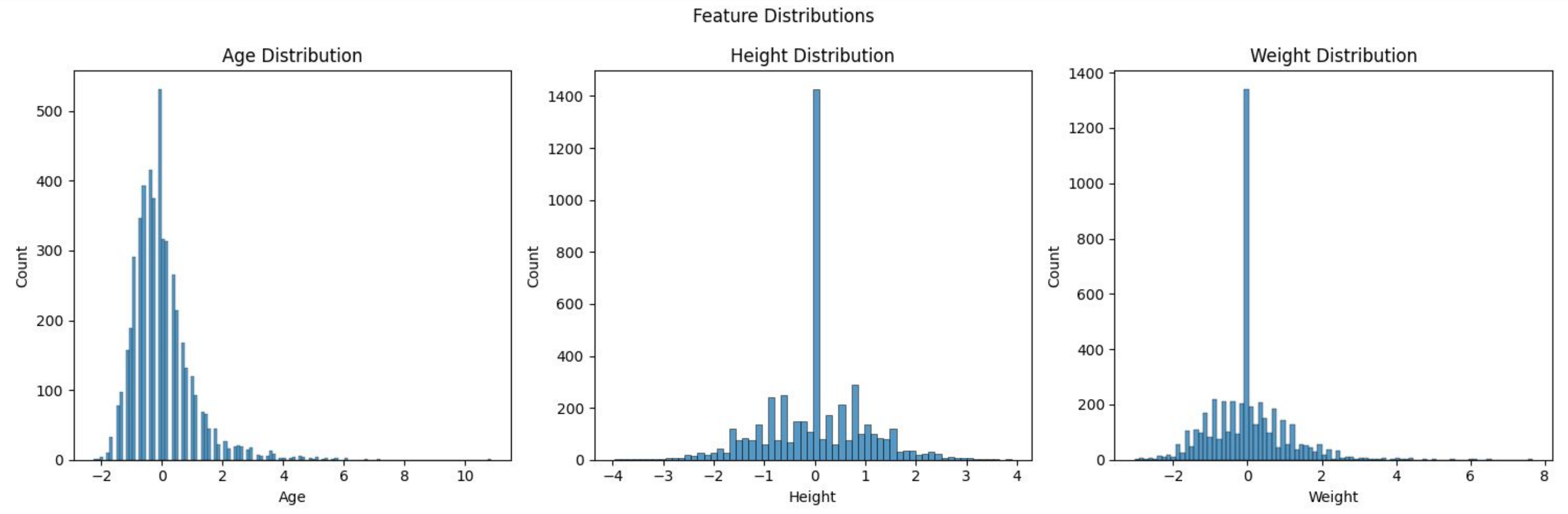
- All three algorithms (Apriori, FP-Growth, and ECLAT) consistently found that medals were evenly distributed at 33.33% for each type (Gold, Silver, Bronze)
- Athletics emerged as the most common sport with 12.03% support
- Swimming and Rowing tied for second place with 6.22% support each

Model Evaluation & Performance Results

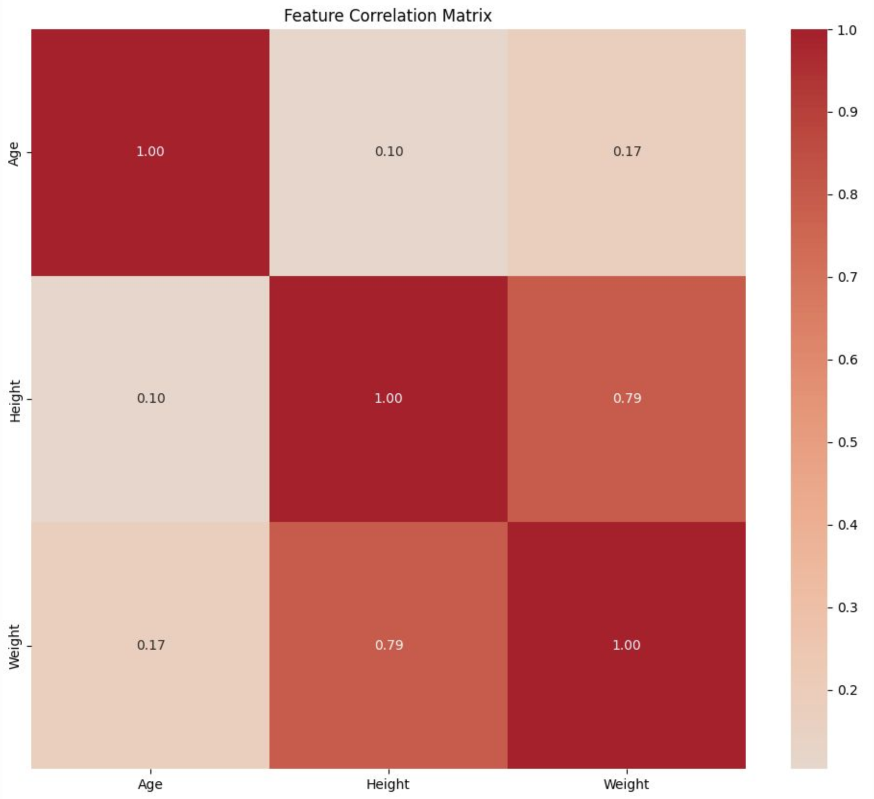
3. Dataset Characteristics:

- The analysis used a balanced dataset of 723 samples from an original size of 4,998
- The preprocessing resulted in 72 features from 9 selected initial features
- All missing values were successfully handled during preprocessing

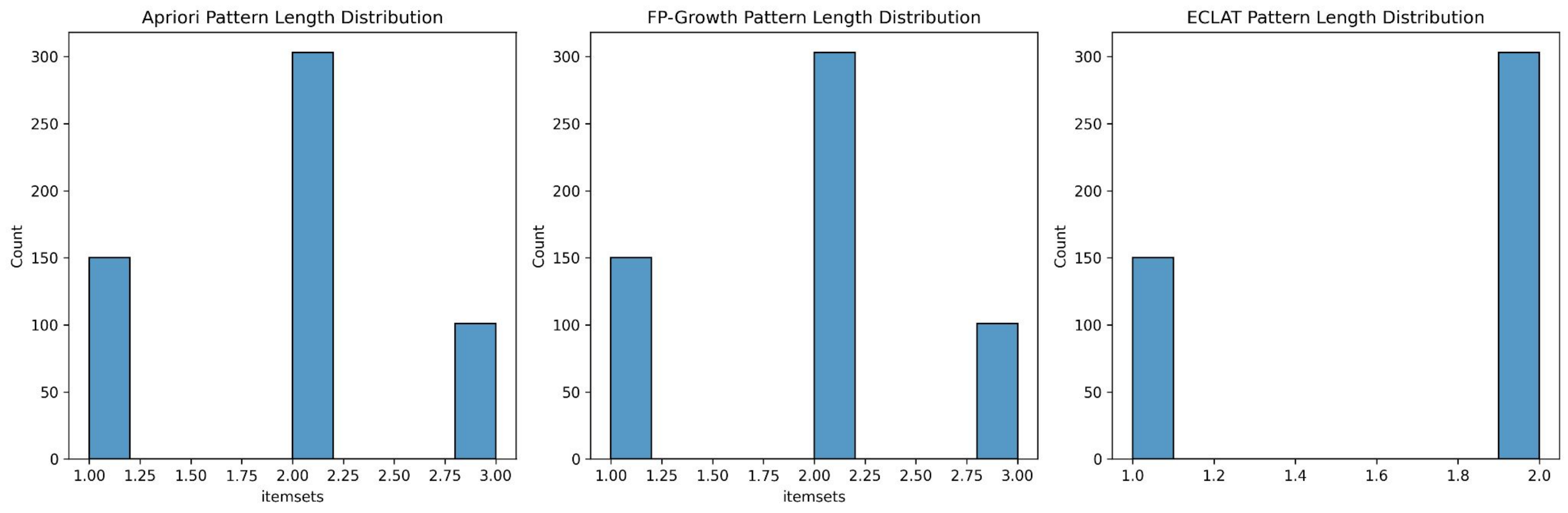
Feature Distribution



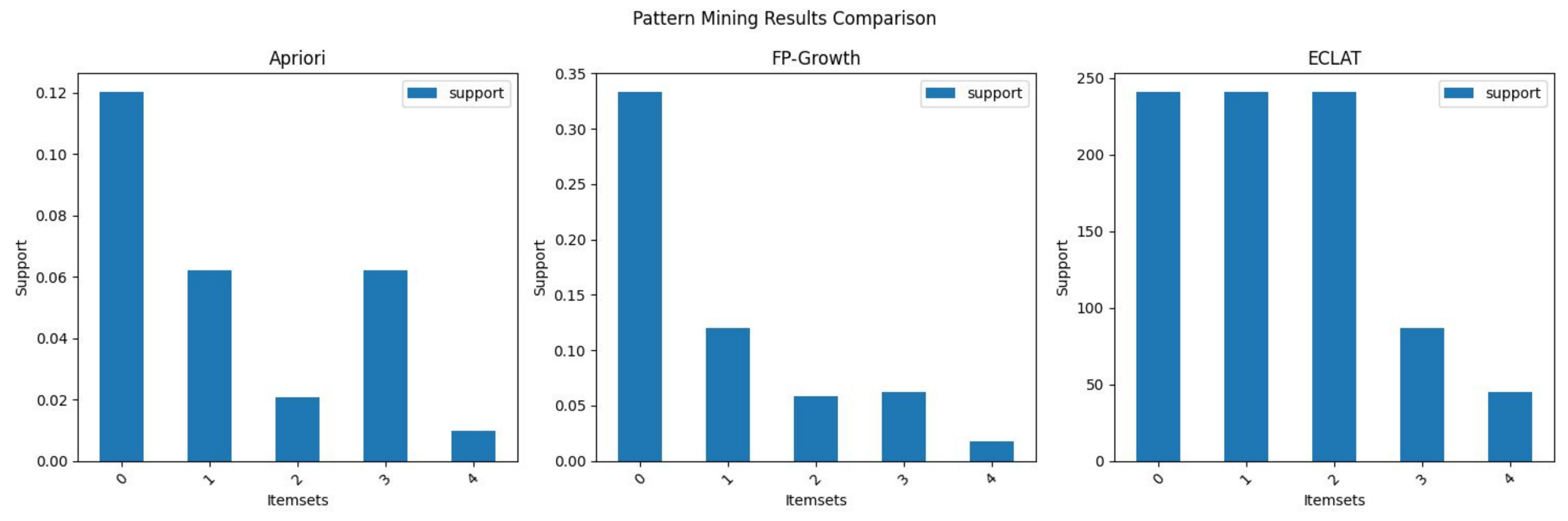
Feature Correlation Matrix



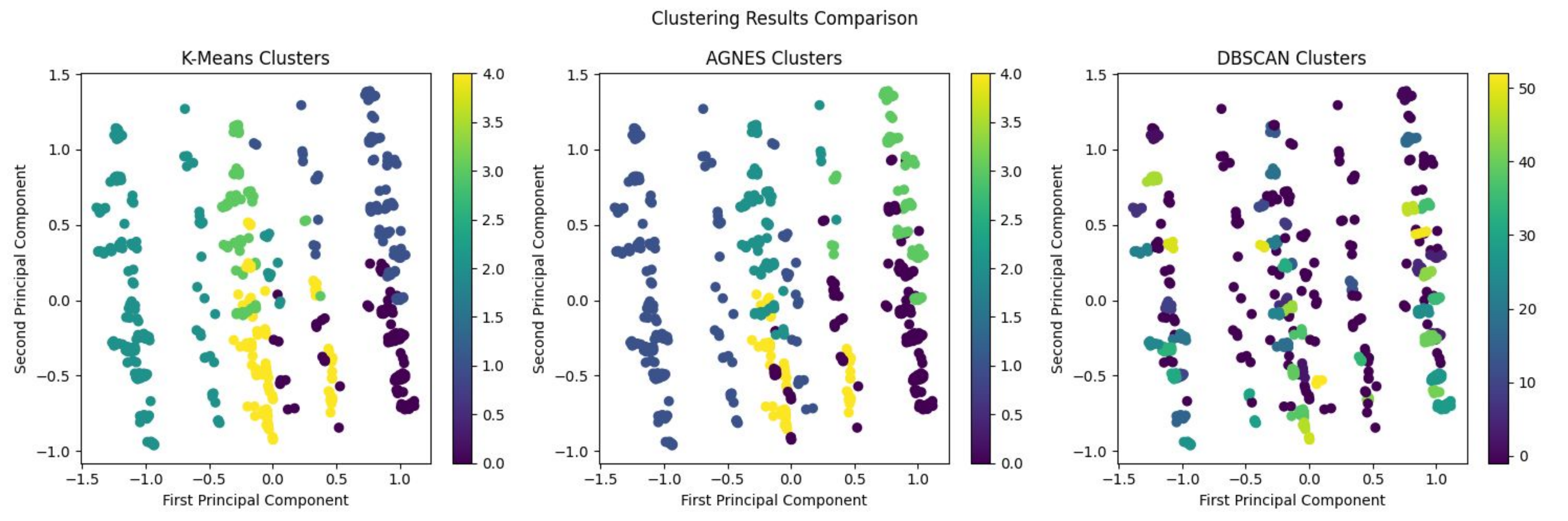
Pattern Analysis



Pattern Mining Results



Clustering Results



Conclusions

Frequent Pattern Mining

Three algorithms (Apriori, FP-Growth, and ECLAT) revealed consistent patterns:

- Equal distribution of medals (33.33% support for each medal type)
- Athletics dominated with 12.03% support
- Swimming and Rowing each showed 6.22% support
- Notable pattern: Athletics-Gold medal combination with 5.81% support

Conclusions

Clustering Analysis

- **K-means (5 clusters):**
  - Moderate separation (Silhouette Score: 0.2703)
  - Notable clusters included:
  - Athletics/Swimming/Rowing cluster with strong Gold medal presence
  - Fencing/Shooting/Equestrianism cluster with balanced medal distribution
- **AGNES (5 clusters):**
  - Showed poorer separation than K-means
  - Similar sport groupings but different medal distributions
- **DBSCAN (53 clusters):**
  - Best separation (Silhouette Score: 0.5462)
  - Created more specialized sport-specific clusters
  - Many small, highly focused clusters (e.g., Gymnastics-only with 7 gold medals)

References

https://www.kaggle.com/code/marcogdepinto/let-s-discover-more-about-the-olympic-games/notebook

Thanks!

Do you have any questions?